Tractor captures R1, R2 and R4

Date: _____ Ambient temp: _____ Battery SoC: _____ %

All three runs are done parked, in neutral with the PTO out, so only the motor and the gearbox input spin. About 15 minutes in all. Write down the clock time at the start of every run: one session file may hold several runs, and the time is how they are found afterwards.

Before you start

- ☐ Logger shown to capture without drops (bench check, then a minute parked in run mode on the tractor). Phone connected so the log has clock time.
- ☐ Zombie on the **gov-logging** firmware. If it was just flashed: set **govFlt = 4**, save, power-cycle (the ID 162 clash otherwise loads 2046 and the governor commands full regen).
- ☐ Tractor stationary: **GovLog** on, then check the MG charger's DC-DC output stays up.
- ☐ Handbrake on, wheels chocked, gearbox in **neutral**, PTO **out**, nobody near the PTO shaft.
- ☐ Gearbox warm (friction depends on oil temperature). Warm-up drive done? yes / no
- ☐ Write down the settings you will change, so they can be put back:
  revlim _____ off-throttle regen setting (name) _____ = _____

R1 No-load acceleration

Gives the inertia J of the motor and gearbox input, and friction at three speeds. 13 Nov 2026 gave J 0.055 kg m2, Tc 4.8 Nm, b 0.045 Nm s/rad: repeat only with a logger proven to keep every frame.

1. Set **revlim = 3000** so the governor stays out of the way.
2. From standstill, press the throttle to about **3 %** (potnom) and hold it still. The shaft is light and friction is nearly flat, so it speeds up and **settles at its own speed** within about 5 s, somewhere between 1500 and 2500 rpm. Do not chase an exact speed: 0.1 % of pedal moves it about 110 rpm. Hold the settled speed for about **5 s**.
3. Release the throttle fully and let it coast to a stop. That coast is R2 (next section), so let it run down without touching anything. It takes about a second.
4. **Abort:** if speed passes **2600 rpm**, release at once.
5. Repeat **three times** at about **2.5, 3 and 3.5 %**. Note the pedal % and the settled rpm.

Run	Clock time	Session file	Pedal %	Settled rpm	Notes
R1 / R2 #1					
R1 / R2 #2					
R1 / R2 #3					

R2 Coast-down

Gives the friction terms Tc, b and c.

1. If practical, set the Zombie's **off-throttle regen to zero** for R1/R2. Otherwise regen torque is mixed in with the friction. If you leave it on, write ON here: _____
2. From about 2000 rpm (the end of each R1), throttle **fully released**, hands off, until the shaft stops. No brake, no clutch, no gear change.
3. R2 is recorded in the same rows as R1 above. Note in the rows if anything disturbed a coast.

R4 Fast hit on the limit

Reference for overshoot above revlim, ringing, and any dip-and-creep. The bench repeats it with the same gains (validation step 2).

Set back to the working governor settings first:

- ☐ **revlim = 2200**
- ☐ Off-throttle regen back to its usual value (written down on page 1)
- ☐ Gains as normal: govKp 11, govKi 0, govKd 0, govImax 100, regenmax -30, throtramp 10, govFlt 4.
  Different? Write here: _____

Expect hunting. In neutral the shaft is so light that these gains cannot settle it: on 13 Nov speed swung about 1600 to 2470 rpm with the pedal held. That is what this run records.

1. Still parked, neutral, PTO out. Start from **standstill**, throttle closed.
2. Step the throttle quickly to about **6 %** (potnom) in one movement, and hold it there. **Never full throttle:** on 13 Nov, 18 % overshot to 3460 rpm against revlim 2200, and full throttle requests about five times that torque.
3. Hold for about **5 s**, then release and let it stop.
4. **Abort:** if speed passes **2700 rpm** (revlim + 500), release at once and note it.
5. Repeat **three times** at the same pedal. Wait for the shaft to stop between repeats.

Run	Clock time	Session file	Pedal %	Peak rpm	Notes
R4 #1					
R4 #2					
R4 #3					

Afterwards

- ☐ revlim back to **2200**, off-throttle regen back to normal. Saved.
- ☐ GovLog: leave on / turn off (circle one). DC-DC still fine?
- ☐ Copy the session files off the logger. Put each run's clock times, pedal % and notes with them (photo of this sheet is fine).
- ☐ On the PC: `python tools/loglib.py session_XXXX.csv` - the 0x1D4 and 0x4A1 gap counts should be near 0, and dropped 0x4A1 frames 0.

What happens to the logs

`extract_params.py` fits J from R1 and Tc, b, c from R2, and writes the simulator settings. The bench then repeats R1 open loop (validation step 1, within 5 %) and R4 with the gains and pedal recorded in the log (step 2: overshoot within 20 % or 30 rpm, whichever is larger, and similar dip-and-creep). Only once those match are bench sweeps trusted for tuning.

Notes